



Diabetic Retinopathy

[What is DR?](#)[Affects of](#)[Screening & Diagnosis](#)[Treatment](#)[Resources](#)

What is Diabetic retinopathy?

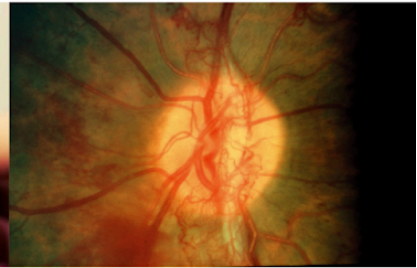
Diabetes is a common disease which leads to several complications, such as microvascular conditions (Wong et al, 2016). Diabetic retinopathy is the leading cause of vision loss, particularly in adults and the elderly (Wong et al, 2016). The progression of the disease can be detrimental without treatment, as new abnormal blood vessels form, leading to the possibility of macular edema (Wong et al, 2016).

Content

1. What is diabetic retinopathy

2. Causes of diabetic retinopathy

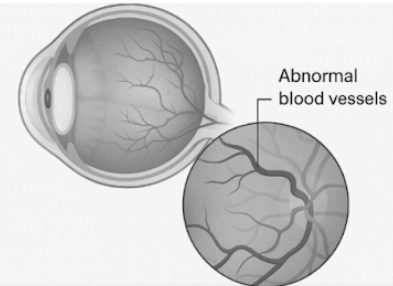
3. Stages of diabetic retinopathy



Causes of Diabetic Retinopathy

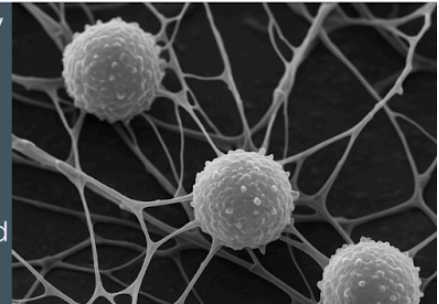
Vascular Changes

Diabetic retinopathy is a microvascular disease which is caused by hyperglycemia (Wang & Lo, 2018). Retinal blood vessels dilate in response to hyperglycemia and change blood flow (Wang & Lo, 2018). An early marker of diabetic retinopathy is the loss of pericytes due to high glucose levels (Wang & Lo, 2018). Pericytes are contractile cells which surround blood vessels, providing structural support and controlling blood flow (Atwell et al, 2015).



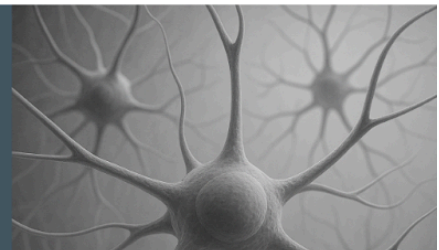
Inflammation

Another factor to consider is inflammation, which has been widely detected marker in diabetic animals and patients (Wang & Lo, 2018). Increased amounts of leukocytes were detected in the retinal vasculature in diabetic animal models, which causes leukostasis where white blood cells start sticking to the walls of capillaries (Wang & Lo, 2018). The adhesion of these cells damages the blood-retina barrier, which normally separates blood components from retinal tissue (Wang & Lo, 2018).

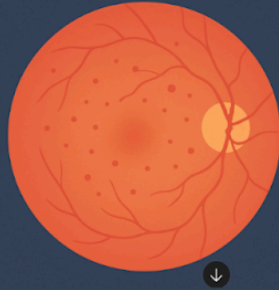


Neuron Death

Retinal neurons can go through the process of apoptosis (programmed cell death), which was shown to happen within a month of diabetes in rats (Wang & Lo, 2018). Proteins such as cytochrome c and other apoptosis factors begin increasing in these neurons, which is likely caused by high glucose exposure



Stages of Diabetic Retinopathy



Mild non-proliferative DR

- Microaneurysms are present in the retina

(Wong et al, 2016)



Moderate non-proliferative DR

- Microaneurysms
- Retinal dots
- Blot haemorrhages

(Wong et al, 2016)



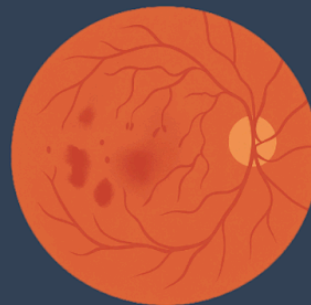
Severe non-proliferative DR

- Intraretinal hemorrhages (more than 20 in each quadrant)

Venous bleeding

Intraretinal microvascular abnormalities

(Wong et al, 2016)

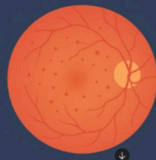


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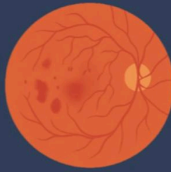
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each card is
its own entity
ex. stage 1, 2 etc

Causes of Diabetic Retinopathy

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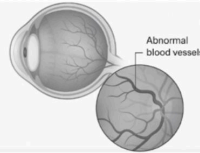
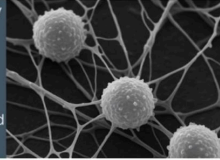


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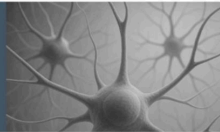
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